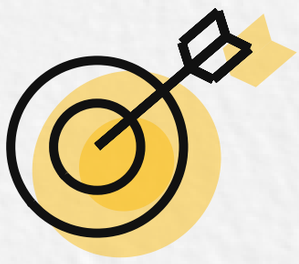
Week 7 - Learning Map



Goal:

Build a hybrid retrieval system that combines dense semantic search and sparse lexical search, uses Qdrant for indexing, fuses rankings with RRF, and integrates the final retriever into an agentic RAG workflow for stronger, more reliable answer generation.

Overview...

Week 7 focuses on improving retrieval quality by combining semantic and lexical search into a hybrid retrieval system.

The module begins with the foundations of dense and sparse retrieval, explaining how dense embeddings capture semantic similarity while sparse methods preserve exact keyword matching.

Learners understand why dense vectors rely on approximate nearest neighbor methods such as HNSW, while sparse retrieval uses inverted indexes for efficient term-based lookup.

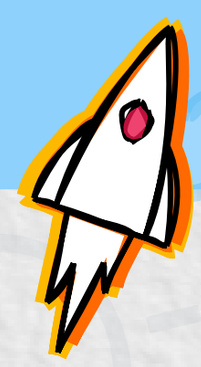
The session then shows why traditional lexical approaches such as TF-IDF and BM25 are still useful, but also where they fail when wording differs between query and document.

To bridge that gap, the module introduces SPLADE as a sparse retrieval method that adds weighting and term expansion.

These ideas come together through hybrid search, where dense and sparse rankings are merged using Reciprocal Rank Fusion.

The notebook makes the theory practical by building a hybrid retriever on an e-commerce dataset using Qdrant, FastEmbed, and pretrained models. The workflow ends by integrating the retriever into an agentic RAG pipeline for stronger, more reliable search-driven answers.

HOW TO START?



Roadmap

Learners are encouraged to follow along with the instructor during the live session and execute the code on their local environment.

Live Class

Retrieval Foundations

- Dense vs sparse retrieval
- HNSW, ANN, and inverted index basics
- Why lexical and semantic search differ

Tool Use + Multi-Agent Orchestration

- BM25 / SPLADE and dense embeddings
- Hybrid retrieval with Reciprocal Rank Fusion (RRF)
- When hybrid search works best

Build the Hybrid Retriever

- Generate dense and sparse embeddings
- Store and query them in Qdrant
- Fuse results and test retrieval quality
- Use hybrid search inside agentic RAG

Assignment

- Build a hybrid search & retrieval system over biomedical research abstracts using the PubMedQA dataset.

Please use the Wednesday Technical Coaching session to clarify any doubts you have on the topics. The assignment solutions will be discussed during the Thursday Assignment Review session.

All the resources have been added to your Uplevel account. If you have any questions, please reach out to the Operations team or raise a support ticket.